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RPC-Bench: A Fine-grained Benchmark for Research Paper Comprehension

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Overview

- Background
- Benchmark Design
- Experimental Results
- Conclusion & Limitations

Motivation

Research paper comprehension is a key prerequisite for reliable research copilots and subsequent scientific discovery.

The screenshot shows a research copilot interface for the paper "Attention is All You Need". On the left, the paper's title and authors are listed. The main area displays the abstract, which describes a new sequence transduction model based on the Transformer architecture. The interface includes a "Paper Details" sidebar with tabs for "AI Summary", "AI Comments", and "note". A "Summary" section provides a concise overview of the paper's contributions. Below the summary, "Recommended Questions" are listed, covering topics like training speed, model interpretability, and performance improvements. A "Full text understood, you can ask questions" message is visible at the bottom.

AI Paper Reading

The screenshot displays the "Hot Topics" Global Large Model Frontier Tracking interface. It features a "Source" section with a list of relevant articles, each accompanied by a small icon and a brief description. A "Dialogue" section on the right provides a detailed overview of the "Hotspot" Global Large Model Frontier Tracking, including its purpose and key findings. A "Tools" sidebar on the far right offers various resources, such as "Core Summary of Academic Ethics and Writing Course", "PrismLLM: Large Model Training Simulation", and "Architecture Design of Real-Time News Stream Processing Agent System in Financial Trading Scenarios". The interface is designed to help researchers stay updated on the latest developments in large language models.

Research Labs

Related Benchmarks

Benchmarks	Papers	QA	Real QA	Taxonomy	Eval. Metrics	Textual inp.	Visual inp.
PeerQA	208	579	✓	task	Corr.	✓	✗
SPIQA	25.5K	270K	✗	task	LLMLogScore	✓	✓
ArXivQA	16.6K	-	✗	task	-	✗	✓
DocGenome	500K	-	✗	task	GPT-acc	✗	✓
RPC-Bench	4150	61.3K	✓	content	Conc., F1-like	✓	✓

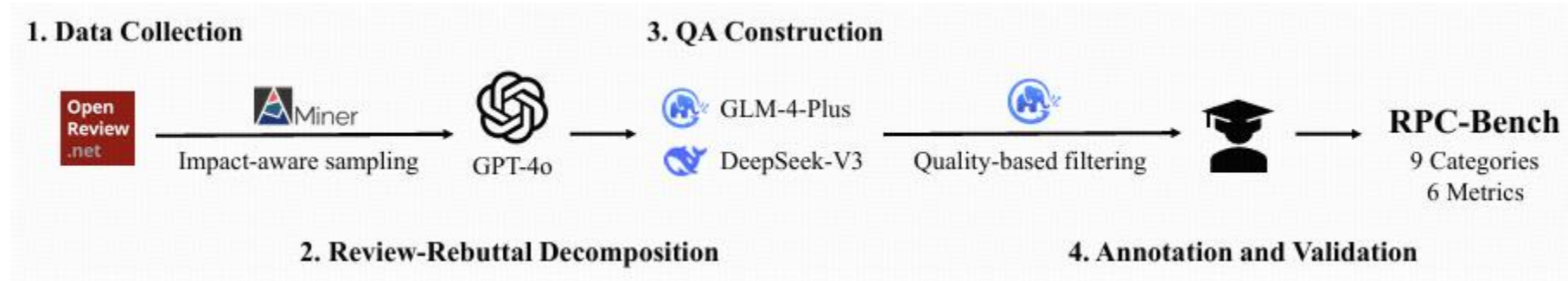
Table 1: Comparison with relevant research paper Benchmarks. Conc.=Conciseness; Corr.=Correctness; F1-like is defined as the harmonic mean of correctness and completeness; inp.=input. “Eval. Metrics” are LLM-based metrics.

- **Limited scale:** few papers and QA pairs
- **Synthetic QA construction:** mostly machine-generated questions
- **Task-oriented taxonomy:** lacks depth-aware categories
- **Narrow evaluation scope:** mainly accuracy-based metrics
- **Limited input formats:** lack parallel text and page-image settings

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Construction Pipeline



✓ Data Collection

- Crawl **papers and review-rebuttal pairs** from OpenReview
- Refine metadata via **AMiner matching**
- Apply **impact-aware sampling** for quality and balance

📄 QA Transformation

- **Decompose** review-rebuttals into comment-response units with GPT-4o
- **Rewrite** units into QA pairs using GLM-4-Plus and DeepSeek-V3
- Organize questions by research-flow taxonomy

⚡ Quality Control

- **Filter** low-quality or paper-unanswerable QA items
- Conduct iterative human annotation and review
- Produce high-quality **paper-grounded** QA pairs

Taxonomy: Natural Research Flow of Academic Papers

Task Taxonomy

1. Concept Understanding (C.U.) [What-4.27%]: Clarifies or explains key concepts, terminology, theoretical viewpoints, or information conveyed in figures, tables, or formulas.
2. Methods
 - 2.1. Method Disambiguation (M.D.) [What-8.91%]: Clarifies methodological details to resolve misunderstandings or ambiguities, ensuring an accurate understanding of proposed approaches.
 - 2.2. Method Mechanics (M.M.) [How-9.91%]: Questions about the implementation or function of methodological workflow or components, such as the effect of specific modules in models.
 - 2.3. Motivation Analysis (M.A.) [Why-6.29%]: Explores the rationale behind a proposed method or decision.
 - 2.4. Method Comparison (M.C.) [11.75%]: Compares the proposed approach with baseline methods, analyzing similarities, differences, or performance to highlight novelty.
3. Experiments
 - 3.1. Experimental Exposition (E.E.) [What-13.07%]: Explains results, infers how changes to experiments could affect results or conclusions, and handling reasoning tasks (i.e., calculations, counting, or comparisons).
 - 3.2. Experimental Setup (E.S.) [How-6.77%]: About the design, configuration, and execution of experiments.
 - 3.3. Experimental Analysis (E.A.) [Why-14.08%]: Studies the reasons of specific experimental results, links them to the proposed approach, and assesses their generalizability and potential impact.
4. Claim Verification (C.V.) [24.95%]: Binary classification tasks that judge whether claims, hypotheses, or experimental conclusions are correct.

Figure 3: Task taxonomy of QA pairs. The form of [What-4.27%] indicates question types and QA percentage.

Annotation Process

Annotation progress
Paper progress: 1/1188
QA progress: 1/20
Overall progress: 1/15195

Paper id: 1
QA id: 1
Overall progress: 1

Guidelines:
1. Specific task: Convert challenging information from English paper review-rebuttals into English QA. LLM has been used for preprocessing and conversion into QA (which may lose some information and is only for reference), but the coverage of information may be incomplete, and experimental tables, formulas, and other information may be lost, requiring manual proofreading and supplementation. In addition, the Chinese translations on the annotation page are

Model 1
Question 1-1: What are the details of the low-pass filter used to address the issue of high-frequency action predictions by the RL policy in Line 176?
Answer 1-1: The RL policy predicts actions at a frequency too high for the PD controller to track. To address this, a butterworth low-pass filter with a high cut frequency of 1.5 is used to produce a smoother action signal. Implementation details are provided in the linked document.
Question 1-1: What are the specific details of the low-pass filter mentioned in line 176 used to address the high-frequency action prediction problem in the RL policy?
Answer 1-1: The RL policy predicts actions at a frequency too high for the PD controller to track. To address this issue, a Butterworth low-pass filter with a high cutoff frequency of 1.5 was used to generate smoother action signals. Implementation details are provided in the linked document.

Model 2
Question 1-1: What are the details of the low-pass filter mentioned in Line 176, including its purpose and implementation?
Answer 1-1: The low-pass filter is used to produce a smoother action signal because the RL policy predicts actions at a frequency that is too high for the PD controller to track. Specifically, a Butterworth filter with a high cut frequency of 1.5 is employed. Further implementation details are available at the provided link.
Question 1-1: What are the specific details of the low-pass filter mentioned in line 176, including its purpose and implementation?
Answer 1-1: The low-pass filter is used to generate smoother action signals because the frequency of actions predicted by the RL policy is too high for the PD controller to track. Specifically, a Butterworth filter with a high cutoff frequency of 1.5 is used. Further implementation details can be found in the provided link.

Annotating data item 1, Not annotated

Model 1
Retain Model 1 1-1 results
Rewrite QA
Review: Line 176 mentioned the use of low-pass filter but the details were not given.
Rebuttal: In the final paper we will explain that the RL policy predicts actions at a frequency that is too high for the PD controller to track. We remedy this issue by using a lowpass filter to produce a smoother action signal. Specifically, we use a butterworth filter with a high cut frequency of 1.5. The implementation details can be found [here](#).

Model 2
Retain Model 2 1-1 results
Review
Line 176 mentioned the use of low-pass filter but the details were not given.
Rebuttal

Overview of Annotation Quality

Total Annotations	Retention Rate	Pass Rate	Pending Review Count
2557	93.7%	13.6%	2135

Filter Criteria
Range Filter (Data Entry index, starting from 1) 1 2557
Global Start Index 1 Global End Index 2557
Review Confidence Filtering: Low, High, Pending Review
Data Annotation Status: Approved, Rejected, Pending Review
Whether Annotation is Retained: Skip, Retain

Review Operation
Pending Review
Review comments (optional for approval, mandatory for rejection)
请输入通过/驳回理由...
Review Confidence: Low, High
Approved, Reject Annotation
Current record: 1/2209

- **LLM candidates:** GLM-4-Plus and DeepSeek-V3 generate taxonomy-based QA candidates.
- **Blind human annotation (left):** annotators choose the better anonymized LLM output, verify labels, revise or split QA pairs, and discard low-quality items.
- **Review loop (right):** reviewers approve or reject annotations with feedback for correction.

Quality Control

Quality Assurance Workflow

Camera-ready grounding

Use final papers as the evidence source, including post-review clarifications, added experiments, and supplementary content.

Two-stage filtering

Remove low-quality or unanswerable items at both comment-response and QA-item levels.

Reviewer validation

Check answerability and quality, return problematic cases for correction, and remove 8.87% low-quality or paper-unanswerable QA items.

Revision source control

Dev/test answers are mainly human-rewritten for factual fidelity..

Quality-Based Filtering Criteria

Temporary or editorial issue

We corrected "benchamrks" to "benchmarks"
Added reference to Smith et al.

External resource dependency

More cases: <https://...>

Non-substantive commitments

We will add a limitations section
Will address in future work

	GLM	DeepSeek	Human
Question	29.20	52.15	18.65
Answer	11.99	22.48	65.53

Table 10: Source distribution (%) of question and answer revisions in the dev/test sets.

Dataset Statistics

Statistics	train	val	test
Papers	3100	850	200
Accept	1980	609	116
year	2013-2021	2022-2023	2024
Venue	12	7	4
QA	45651	12895	2787
A/M Q	25.2/105	26.2/261	24.2/250
A/M A	70.5/297	121.5/1337	87.9/773

Table 2: Statistics of the RPC-Bench. A/M Q: average/max question length. A/M A: average/max answer length. Lengths are measured in words.

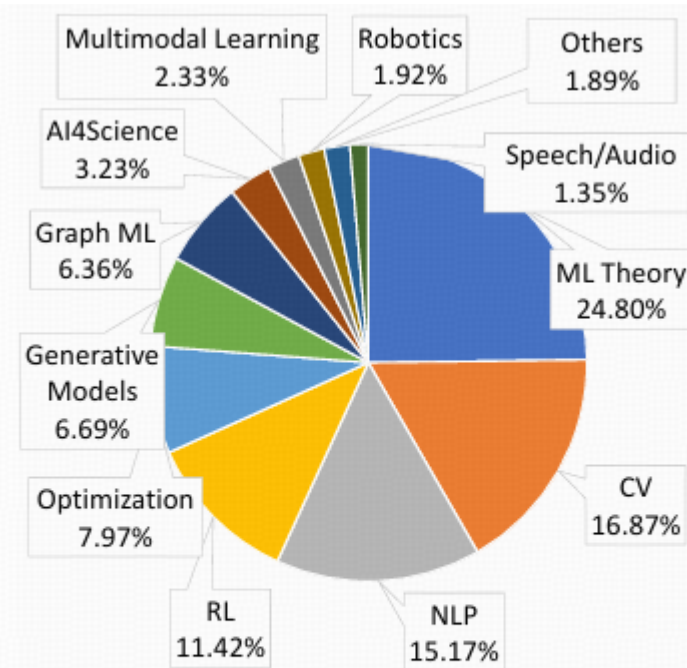


Figure 2: Domain distribution of RPC-Bench. ML: Machine Learning; CV: Computer Vision; NLP: Natural Language Processing; RL: Reinforcement Learning.

Evaluation Protocol

Claim verification: accuracy for binary True/False questions

Open-ended QA: LLM-as-a-Judge scoring for long-form answers

Three Core Dimensions:

- **Correctness:** factual accuracy and faithfulness
- **Completeness:** coverage of essential information
- **Conciseness:** brief, focused, and non-redundant answers

Each dimension is rated on a 0-5 scale

$$\text{F1-like} = \frac{(1 + \beta^2) \times (\text{Correctness} \times \text{Completeness})}{\beta^2 \times \text{Correctness} + \text{Completeness}}$$

Harmonic mean of correctness and completeness

$$\text{Informativeness} = \text{F1-like} \times \frac{\text{Conciseness}}{5}$$

Combines all three dimensions and penalizes verbose outputs

LLM Judgments vs. Human Assessments

Setting	P-BT	S-BT	PW-AUC	Avg.
AUG+SEP	0.8955	0.9137	0.7125	0.8406
AUG+JOI	0.9003	0.9091	0.6966	<u>0.8353</u>
RAW+SEP	0.8773	0.9137	0.7054	0.8321
RAW+JOI	0.8759	0.9137	0.7100	0.8332

Table 4: Agreement between human and model judgment w.r.t. prompt configurations. (AUG: enhancement with title and abstract; SEP/JOI: separate/joint evaluation)

Model	P-BT	S-BT	PW-AUC	Avg.
GPT-5 _{int}	0.9059	0.9227	0.7027	<u>0.8437</u>
Claude-4.5 _{int}	0.8928	0.9136	0.7038	0.8367
Gemini-3 _{int}	0.9091	0.9227	0.7090	0.8469
GPT-5 _{dec}	0.9213	0.9137	0.7255	<u>0.8535</u>
Claude-4.5 _{dec}	0.8873	0.9091	0.7197	0.8387
Gemini-3 _{dec}	0.9170	0.9182	0.7330	0.8561

Table 5: Agreement between different LLM judges and human assessments. Subscript *int* denotes integer scoring, and subscript *dec* denotes decimal scoring.

Human Alignment Study

Sample 300 open-ended QA instances and collect predictions from all evaluated models.

Build **length-controlled pairwise comparisons** to reduce answer-length bias.

Human annotators select the more correct and complete answer under **randomized order** and **masked model identities**.

Measure alignment using **Bradley-Terry correlation** and **pairwise AUC**.

Human-Aligned Judge Configuration

Compare prompt settings, judge models, and scoring schemes

The most human-aligned setup:

- **title/abstract-augmented prompts**
- **separate scoring for each dimension**
- **decimal scores**
- **averaged judgments from GPT-5 and Gemini-3-pro**

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Experimental Setup

28 models (**LLMs, VLMs, Document-Centric Models, RAG Models**) evaluated on the RPC-Bench test set

- **Text-only:** PDFs parsed into Markdown with MinerU
- **Image-based:** PDFs rendered as page images at 200 DPI, (V) denotes image-based input

Inference & Evaluation

- Answers must rely only on the given paper.
- **Open-ended QA:** concise responses under 3,000 characters; evaluated by ROUGE-L, BERTScore, correctness, completeness, conciseness, F1-like, and informativeness.
- **Claim verification:** strictly True/False outputs; evaluated by accuracy.
- All scores are normalized to a 0-100 scale.

Main Results

Model Type	Model	Traditional		LLM-as-a-judge				
		R-L	BERTS.	Concise.	Correct.	Complete.	F1-like	Info.
LLM	DeepSeek-V3.2	19.22	<u>55.60</u>	56.31	58.73	55.19	56.91	32.04
	GLM-4.7	17.09	48.58	54.34	54.36	51.75	53.02	28.81
	Qwen3	16.16	54.25	41.44	55.88	56.64	56.26	23.31
	GPT-5	16.89	54.52	54.93	69.10	67.33	68.20	37.46
	Claude-4	16.60	54.02	41.37	58.53	58.44	58.48	24.19
	Gemini-2.5	18.24	55.67	54.87	<u>62.65</u>	<u>59.03</u>	<u>60.79</u>	33.35
DCM	DocOwl2(V)	14.32	46.42	50.19	11.75	6.66	8.50	4.27
	Docopilot(V)	16.92	53.82	39.31	18.31	17.12	17.69	6.96
	Monkey(V)	20.16	55.19	54.61	17.08	11.27	13.58	7.41
VLM	GLM-4.6V	<u>19.38</u>	54.76	64.55	47.32	43.43	45.29	29.23
	Qwen3(V)	14.70	53.72	22.64	20.17	20.14	20.16	4.56
	GPT-5(V)	17.32	54.85	61.47	58.90	55.34	57.07	<u>35.08</u>
	Claude-4(V)	13.33	50.63	31.63	54.16	53.32	53.74	16.99
	Gemini-2.5(V)	17.27	54.85	51.71	48.39	45.59	46.95	24.28
RAG	HippoRAG2	18.71	54.16	45.77	33.13	27.88	30.28	13.86
	MemoRAG	13.55	52.70	51.31	24.19	19.10	21.35	10.96
	Raptor	18.35	54.00	36.47	25.28	20.82	22.84	8.33
	VdocRAG(V)	17.77	52.22	<u>61.54</u>	21.17	13.88	16.77	10.32
	VisRAG(V)	16.80	54.93	39.90	26.24	23.63	24.87	9.92

Table 3: Evaluation results of free-form QA on the test set. R-L=ROUGE-L; BERTS.=BERTScore; Concise.=Conciseness.; Correct.=Correctness; Complete. = Completeness; Info. = Informativeness. The best results are highlighted in **bold**, and the second-best results are underlined.

Performance across Task Categories

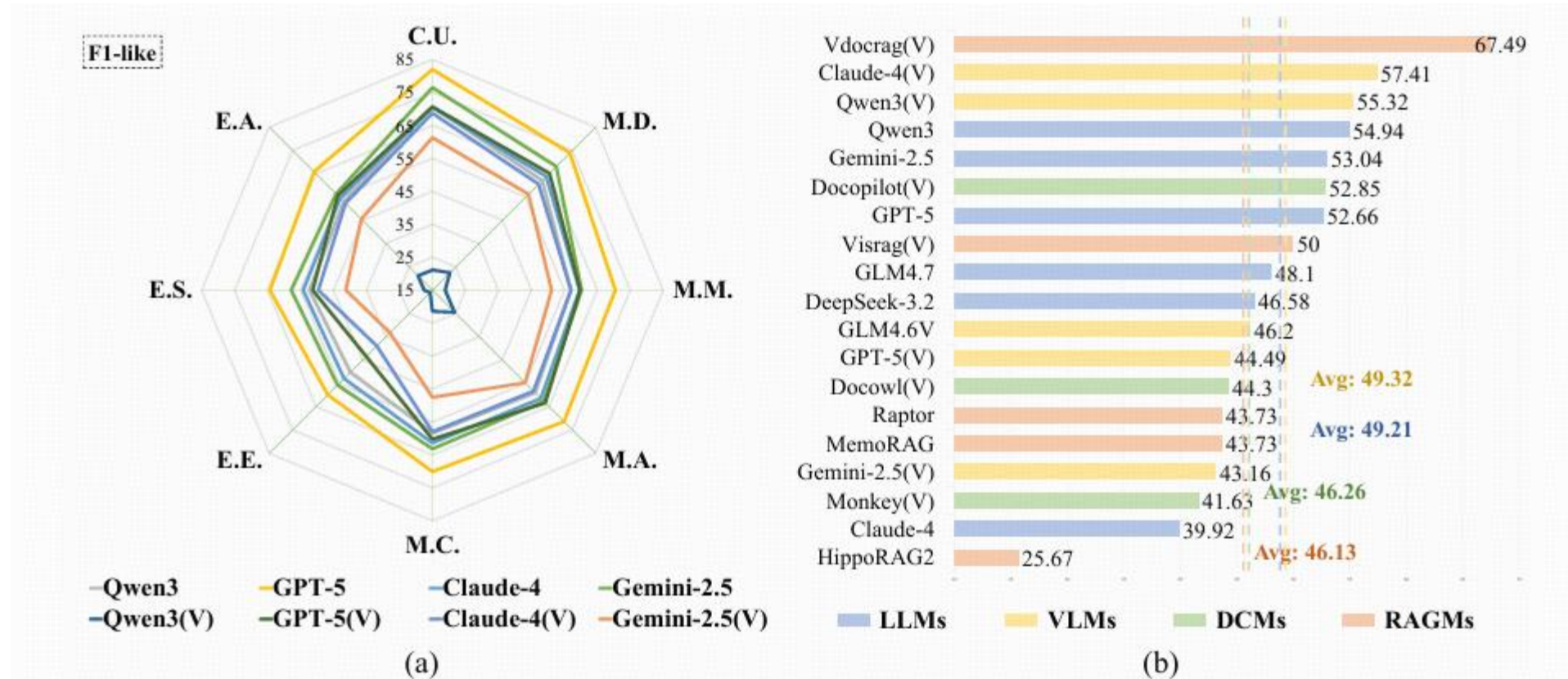


Figure 4: Comparison of LLMs and VLMs on open-ended question answering (F1-like score; left), and the performance of all models on claim verification tasks (ACC; right).

Case Study

Example 1

Paper: NaturalSpeech 2: Latent Diffusion Models are Natural and Zero-Shot Speech and Singing Synthesizers

Question: What are the roles of SoundStream and WaveNet ... and how do they interact with other components such as the prior model, diffusion model, and audio codec?

Answer: SoundStream is used as the neural audio codec ... WaveNet is chosen as the architecture for the diffusion model ... The system consists of three components: a prior model, a diffusion model, and an audio codec...

VdocRAG(V): ...The audio codec and the diffusion model interact with the audio decoder to obtain the audio waveform. The audio codec ... audio waveform. [The previous sentence is repeated over 100 times, resulting in a 19,867-character response].

Example 3

Paper: Safe RLHF: Safe Reinforcement Learning from Human Feedback

Question: What were the experimental results comparing the paper's method to additional safety RLHF baselines such as Constitutional AI and Safety SFT?

Answer: Win rates for three different methods rated by GPT-4: ... — Constitutional AI — Safety SFT — — Helpfulness Win Rate — 71.8% — 40.2% — 53.6% — ... the experiments reveal interesting insights...

Claude-4: ...the experimental comparison of Safe RLHF to Constitutional AI and Safety SFT baselines was very limited. The paper did not conduct comprehensive experiments comparing their method to Constitutional AI and Safety SFT.

Example 2

Paper: AnimateDiff: Animate Your Personalized Text-to-Image Diffusion Models without Specific Tuning

Question: What are the visualization results when the domain adapter training (stage 1) is entirely omitted from the process...?

Answer: The visualizations show that when the domain adapter is entirely removed from the training pipeline, the watermarks from the training dataset appear on the synthetic animations... These results are presented in Appendix B.2 and Fig.10.

GPT-5: Omitting domain adapter training ... This occurs because visual appearance and motion are entangled... (Appendix B.2, Fig. 10, 1st row).

GPT-5(V): When the domain adapter training (stage 1) is omitted entirely, ... resulting in artifacts such as visible watermarks transferring into the generated visuals (Figure 10).

Example 4

Paper: Eureka: Human-Level Reward Design via Coding Large Language Models

Question: Does the EUREKA generate reward functions statistically significantly better than human-designed rewards, based on a Mann-Whitney U-Test with a p-value threshold of 0.05?

Answer: True

Claude-4: FalseTrue

Claude-4(V): True1

Degenerative output:
Repetitive and uninformative responses

Missing multimodal grounding:
Text-only reasoning misses visual evidence

Hallucination: Incorrectly denies evidence present in the paper

Format violation: Fails to follow required True/False output

Case Study

Advanced LLMs: Stronger at contextual integration, method explanation, and experiment reasoning.

VLMs: Useful for figure/table-related evidence, but often **miss fine-grained textual details**.

RAG Systems: Improve factual retrieval, but **struggle with question understanding and multi-source integration**.

DCMs: Often produce **generic, repetitive, or incomplete answers** on more complex questions.

Text inputs usually produce more accurate and complete answers than page-image inputs.

Visual inputs can help with high-level scope or visual evidence, but struggle with **dense PDF layouts**.

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Conclusion & Limitations

Conclusion

RPC-Bench provides a large-scale testbed with **4,150** papers, **61.3K** QA pairs, and **9** categories covering the natural research flow of academic papers.

The benchmark is built from **authentic review-rebuttal exchanges**, using LLMs for scalable QA transformation and humans for quality supervision.

The evaluation protocol measures long-form answer quality, jointly assessing correctness, completeness, conciseness, informativeness, and claim verification.

Experiments on **28** state-of-the-art models reveal clear limitations, with even GPT-5 reaching only **68.2%** F1-like and **37.46%** informativeness.

Limitations & Future Directions

Training data is partly LLM-reformulated

Due to annotation cost, human verification is prioritized for **dev/test**, while train data is released for flexible use and refinement.

Domain coverage is currently CS-centered

The benchmark mainly reflects OpenReview **computer science papers**, though the pipeline can extend to other disciplines with review-response data.

Evaluation focuses on single-paper understanding

Future extensions can target **cross-paper reasoning**, literature synthesis, and broader scholarly workflows.

Project Page

<https://rpc-bench.github.io/>